

EDWIN CALLES MOLINA

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U.S. Citizen – Eligible for U.S. Government Security Clearance | Portfolio: edwincallesmolina.github.io

OBJECTIVE

Aerospace Engineering student seeking a Summer 2027 internship in aerospace design, structures, propulsion, testing, or manufacturing.

EDUCATION

California State Polytechnic University, Pomona

Expected May 2029

Bachelor of Science, Aerospace Engineering | GPA: 3.97

RELEVANT COURSEWORK

Systems Engineering & Design, Aerospace Computational Methods, Engineering Statics, Materials Science & Engineering, Engineering Graphics (SolidWorks), Advanced Engineering Mathematics, Air & Rocket Propulsion Labs

PROJECTS & EXPERIENCE

Particle Image Velocimetry (PIV) Club, Pomona, CA

August 2026 – Present

Design Team Systems Liaison

- Conducted an air-supply trade study for leading-edge active air flow control blowing, showing internal fans would need 1.3–3.1 kW at 250 ft/s versus the <100 W they provide, which justified an external pressurized supply.
- Built a parametric Excel model sizing the blowing holes and supply pressure across freestream velocities of 80–250 ft/s and angles of attack from 0° to stall, serving as the design's numerical proof of concept.
- Verified the project's primary supporting source and caught a m/s-to-ft/s misreading, correcting its tested range from 10–60 to 33–213 ft/s against the 100–200 ft/s airspeed goal.

Bronco Launch – Cal Poly Pomona High Power Rocketry, Pomona, CA

March 2026 – Present

Bronco Avionics – Co-Lead, Launch Operations

- Led the avionics for EXO Bronco, a two-stage rocket targeting the edge of space.
- Set up the sub-team's documentation structure and naming convention, and wrote the next vehicle's avionics bay design constraints.
- Traced a lost booster deployment-channel continuity to corroded screw terminals, replaced them, and led the two-stage pop test that fired both stages correctly.

Independent Design Project – Dual-Deploy High Power Rockets, Ontario, CA

June 2026 – August 2026

- Designed, built, and flew two dual-deploy high power rockets across four flight tests, earning National Association of Rocketry (NAR) Level 2 certification on a 4,135 ft flight.
- Designed and 3D-printed the onboard electronics bays, which protected the recovery electronics through launch, ejection, and landing on every flight.
- Test-flew the second vehicle to 2,833 ft, then flew it supersonic to Mach 1.2 (1,367 ft/s), where it broke apart in flight; recovered the electronics and extracted the full flight data.
- Analyzed the flight data and recovered wreckage to determine that fin failure at supersonic speed caused the breakup, then specified a stiffer fin and composite airframe redesign.

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA), Pomona, CA September 2025 – December 2025

Sub-Team Lead, Small Rocket Competition (SRC)

- Directed a 4-person team across research and design assignments.
- Set the airframe and fin configuration against a 2,800 ft target altitude, checked the stability margin in OpenRocket, and approved the computer-aided design (CAD) before printing.
- Produced and reviewed 3D-printed prototype parts, revising the design based on fit and test results.
- Built and programmed an onboard system that recorded temperature and altitude in flight.

SKILLS

CAD: SolidWorks (part modeling, assemblies, drawings); Simulation: OpenRocket flight simulation; Programming: MATLAB, C++, Java, embedded sensor and data-logging code; Manufacturing: Fused Deposition Modeling (FDM) 3D printing / additive manufacturing, soldering, wiring harness assembly; Analysis: flight data analysis, failure investigation, trade studies, requirements documentation, technical reporting; Microsoft Office: Excel, Word, Gantt charts

HONORS, AWARDS & CERTIFICATIONS

NAR Level 2 High Power Rocketry Certification (August 2026); NAR Level 1 Certification (Spring 2026); President's Honor List (2025–2026)

PROFESSIONAL ORGANIZATIONS

National Association of Rocketry (NAR) – Member, 2026 – Present